LofreeCP

LofreeCP[91]

- **Methodology：**
  - **Frequency**
  - **NE**
  - **SS**
- **Steps:**
  - **Frequency：**

$$F(\hat{y}_a^{(i)}, m) = \frac{\tilde{p}[\hat{y}_a^{(i)}]}{m},$$

  - **NE：**

$$H(x^{(i)}|\{\hat{y}_j^{(i)}\}_{j=1}^m) = |\frac{\sum_{a=1}^n \tilde{F}(\hat{y}_a^{(i)}) \log(\tilde{F}(\hat{y}_a^{(i)}))}{\log m}|$$

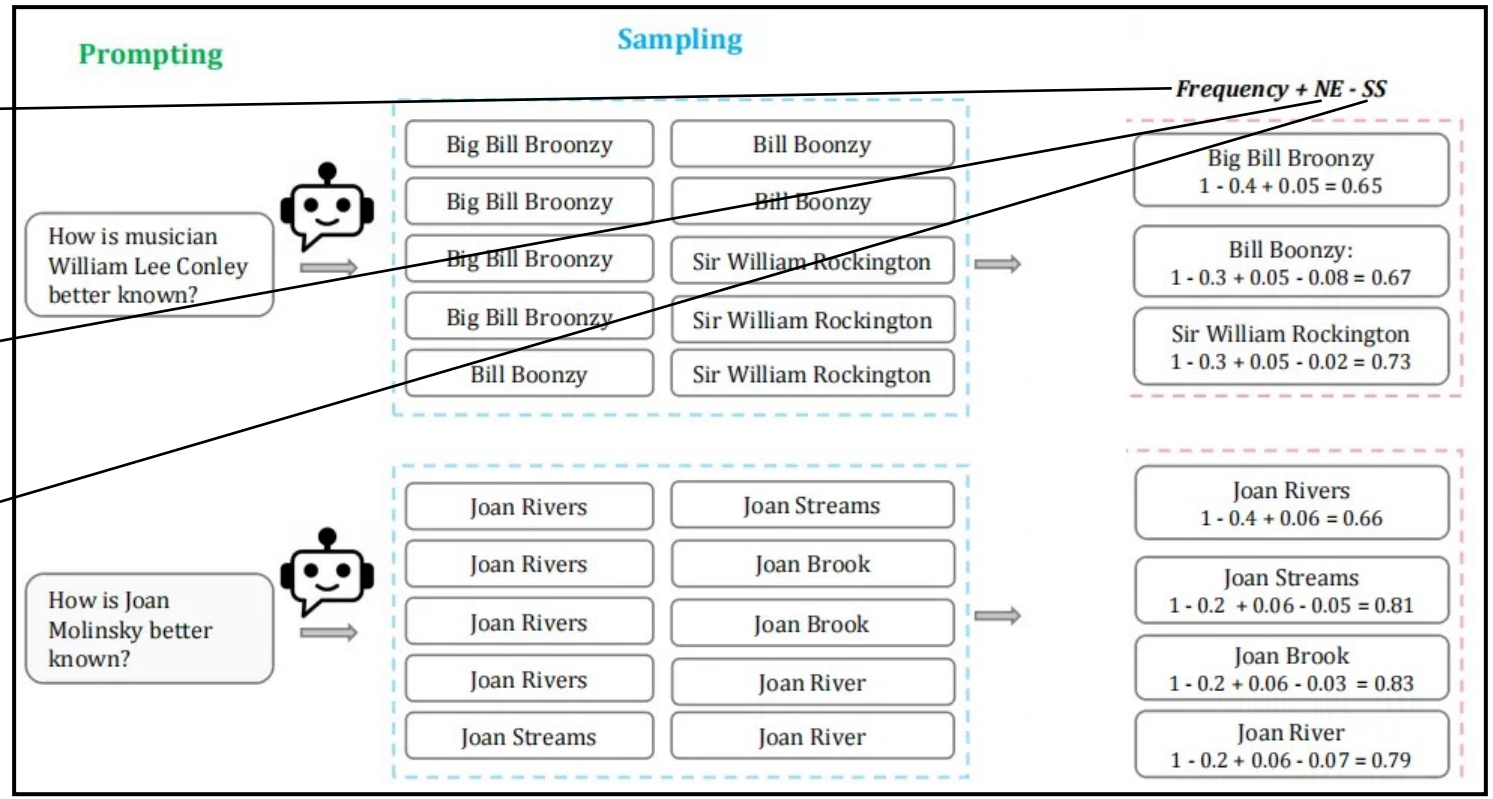
  - **SS：**

$$SS(\hat{y}_a^{(i)}, P_{\text{highest}}^{(i)}) = \frac{\mathbf{v}(\hat{y}_a^{(i)}) \cdot \mathbf{v}(P_{\text{highest}}^{(i)})}{\|\mathbf{v}(\hat{y}_a^{(i)})\| \cdot \|\mathbf{v}(P_{\text{highest}}^{(i)})\|}$$

The same as S（X，Y）

$$N^{(i)} = -F(\hat{y}_a^{(i)}, m) + \lambda_1 \cdot H(x^{(i)}|\{\hat{y}_j^{(i)}\}_{j=1}^m) - \lambda_2 \cdot SS(\hat{y}_a^{(i)}, P_{\text{highest}}^{(i)}),$$

Frequency　NE　SS



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- **Task Type:** Open-ended generation

- **Introduction:** LofreeCP builds upon the CP method by transforming the computation of softmax into the computation of frequency, semantic entropy, and semantic similarity, thus providing a computation method for black-box models that does not require obtaining logits.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Consensus-based	Decoding	Black-box	Post-hoc	Medium	Coverage/Ranking

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